

Project Report

Detect Sleep States

Detect sleep onset and wake from wrist-worn accelerometer data

Submitted by:

Harsheen Kaur (102317037)

Harshita Bisht (102317053)

Submitted to:

Dr. Manshi Sharma



Computer Science and Engineering Department

Thapar Institute of Engineering & Technology, Patiala Jan-June 2025

1. INTRODUCTION

Sleep is a fundamental biological process essential for maintaining cognitive function, emotional stability, and physical health. In children and adults alike, irregular sleep patterns are strongly associated with reduced productivity, impaired memory, and long-term health complications. Traditional sleep monitoring techniques such as polysomnography (PSG), although highly accurate, are expensive, intrusive, and impractical for continuous large-scale monitoring.

With the rapid advancement of wearable technology, accelerometer-based sleep tracking has emerged as a promising alternative. These devices capture continuous motion signals, enabling non-invasive monitoring of sleep patterns in real-world environments. However, extracting meaningful sleep events such as sleep onset and wake-up transitions from raw accelerometer data remains a challenging task due to noise, inter-subject variability, and severe class imbalance.

To address these challenges, this project proposes a deep learning-based approach for sleep state detection using time-series accelerometer data. Specifically, a hybrid Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) architecture is employed to capture both spatial and temporal dependencies in the data. The proposed system aims to provide an automated, scalable, and accurate solution for sleep monitoring.

This report presents the methodology, implementation, and evaluation of the proposed model, along with a comparative analysis against a baseline CNN model.

2. LITERATURE REVIEW

Recent years have witnessed significant advancements in sleep monitoring using wearable sensor data. Early approaches relied on traditional machine learning algorithms such as Decision Trees, Random Forests, and Support Vector Machines (SVMs). While these methods performed reasonably well on static datasets, they lacked the ability to model temporal dependencies inherent in time-series data.

The introduction of deep learning techniques revolutionized this domain. Convolutional Neural Networks (CNNs), initially developed for image processing tasks, have been effectively adapted for time-series analysis. CNNs are capable of extracting local patterns and features from raw accelerometer signals, making them suitable for detecting micro-level activity changes.

Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) networks, have demonstrated strong performance in sequential data modeling. LSTMs address the vanishing gradient problem and are capable of learning long-term dependencies, which are crucial for identifying transitions between sleep states.

Several studies have shown that hybrid architectures combining CNN and LSTM outperform standalone models. CNN layers act as feature extractors, while LSTM layers capture temporal dependencies across time windows. This combination has been successfully applied in activity recognition, healthcare monitoring, and sleep analysis.

To address class imbalance, which is a common issue in event detection problems, Lin et al. (2017) introduced Focal Loss. This loss function reduces the impact of easily classified samples and focuses more on hard-to-classify minority classes, improving model robustness.

Recent advancements (2023–2024) include the use of attention mechanisms and transformer-based models for time-series analysis. While these models achieve high accuracy, they require large datasets and significant computational resources. Therefore, for practical and scalable applications, CNN-LSTM hybrids remain a balanced and efficient choice.

3. METHODOLOGY

3.1 Model Architecture

This project implements two models:

- **Baseline Model:** Convolutional Neural Network (CNN)
- **Proposed Model:** Hybrid CNN + LSTM

The CNN model focuses on extracting spatial features from accelerometer signals, while the hybrid model enhances this by incorporating temporal learning through LSTM layers.

CNN Architecture

- Conv1D (32 filters)
- Conv1D (64 filters)
- Flatten
- Dense (64 units)
- Dropout (0.3)
- Output Layer (Softmax)

CNN + LSTM Architecture

- Conv1D (32 filters)
- Conv1D (64 filters)
- LSTM (64 units)
- Dense (64 units)
- Dropout (0.3)
- Output Layer (Softmax)

3.2 Activation Functions

- **ReLU (Rectified Linear Unit):** Used in hidden layers for non-linearity
- **Softmax:** Used in the output layer for multi-class classification

3.3 Loss Function

- **CNN:** Sparse Categorical Crossentropy
- **CNN + LSTM:** Focal Loss (to handle class imbalance effectively)

3.4 Optimizer

- **Adam Optimizer** is used due to its adaptive learning rate and faster convergence.

3.5 Workflow

The overall pipeline of the system is as follows:

Data Collection → Preprocessing → Labeling → Windowing → Training → Validation → Prediction → Post-processing

Detailed Steps:

1. **Data Collection:**
Accelerometer features such as *anglez* and *enmo* are used.
2. **Preprocessing:**
 - Handling missing values
 - Normalization of data
3. **Labeling:**
 - 0 → Normal
 - 1 → Sleep Onset
 - 2 → Wake
4. **Windowing:**
 - Window Size = 200
 - Stride = 100
5. **Training:**
 - Data split into training and validation sets
6. **Prediction:**
 - Model predicts sleep states
7. **Post-processing:**
 - Clustering nearby predictions to remove duplicate detections

3.6 Implementation Details (Hyperparameters)

Parameter	Value
Window Size	200
Stride	100
Event Window	± 50
Batch Size	64
Epochs	5
Dropout	0.3
Optimizer	Adam
Loss Function	Cross entropy / Focal Loss

4. RESULTS AND ANALYSIS

4.1 Evaluation Metrics

The performance of the models is evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score

4.2 Results Comparison

Model	Accuracy	Precision	Recall	F1 Score
CNN	0.79	0.62	0.79	0.69
CNN + LSTM	0.79	0.72	0.79	0.72

4.3 Analysis

The baseline CNN model achieved reasonable accuracy but failed to effectively detect minority classes (sleep onset and wake events). This is primarily due to class imbalance, where the majority of the data belongs to the normal state.

The hybrid CNN-LSTM model showed improved precision and F1-score, indicating better detection of critical sleep events. The LSTM layer successfully captured temporal dependencies, which are essential for modeling transitions between sleep states.

4.4 Confusion Matrix Analysis

The confusion matrices presented in Fig. 1 and Fig. 2 highlight the performance differences between the CNN and CNN-LSTM models in classifying sleep states.

As observed in Fig. 1, the CNN model predominantly predicts class 0 (normal state), with all samples being classified into this category. While this results in a high number of correct predictions for the majority class, it completely fails to identify the minority classes, namely sleep onset (class 1) and wake events (class 2). This behavior indicates a strong bias toward the dominant class, primarily due to class imbalance and the model's inability to capture temporal dependencies in the data.

In contrast, the confusion matrix in Fig. 2 demonstrates that the CNN-LSTM model achieves improved classification performance across all classes. Although the model still shows a preference for the majority class, it is able to correctly predict a portion of the minority class instances. This improvement is attributed to the LSTM layer, which effectively captures temporal patterns and transitions in the sequential accelerometer data.

Furthermore, the reduction in misclassification of sleep onset and wake events indicates that the hybrid model is more capable of distinguishing between different sleep states. However, some degree of class imbalance still persists, as reflected by the higher number of predictions for class 0 compared to other classes.

Overall, the comparison demonstrates that while the CNN model suffers from severe bias and underfitting, the CNN-LSTM model significantly enhances the detection of minority classes, making it more suitable for sleep event prediction tasks.

- CNN model predominantly predicts class 0 (normal state)

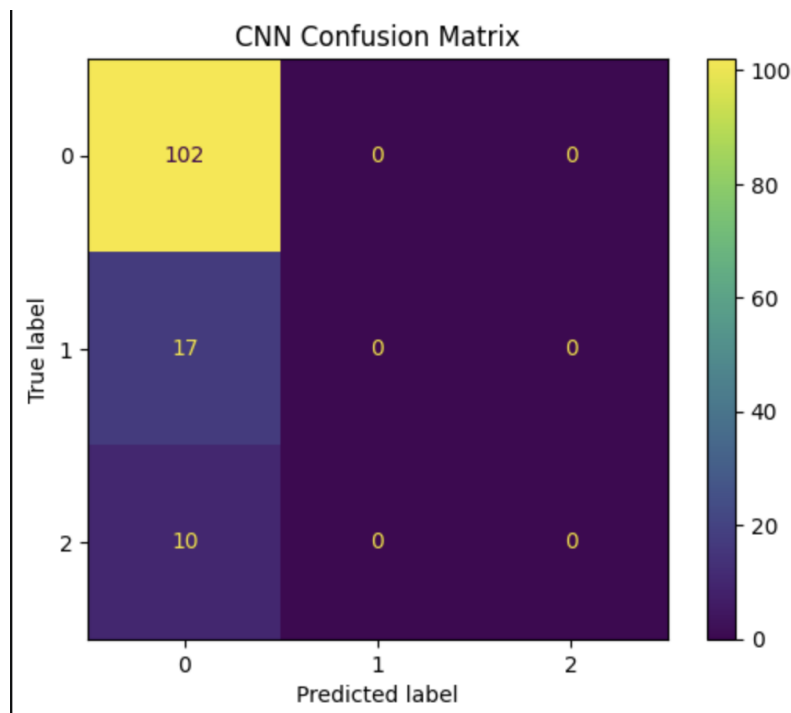


Fig. 1. Confusion matrix of the CNN model for sleep state classification.

- CNN + LSTM improves classification of minority classes

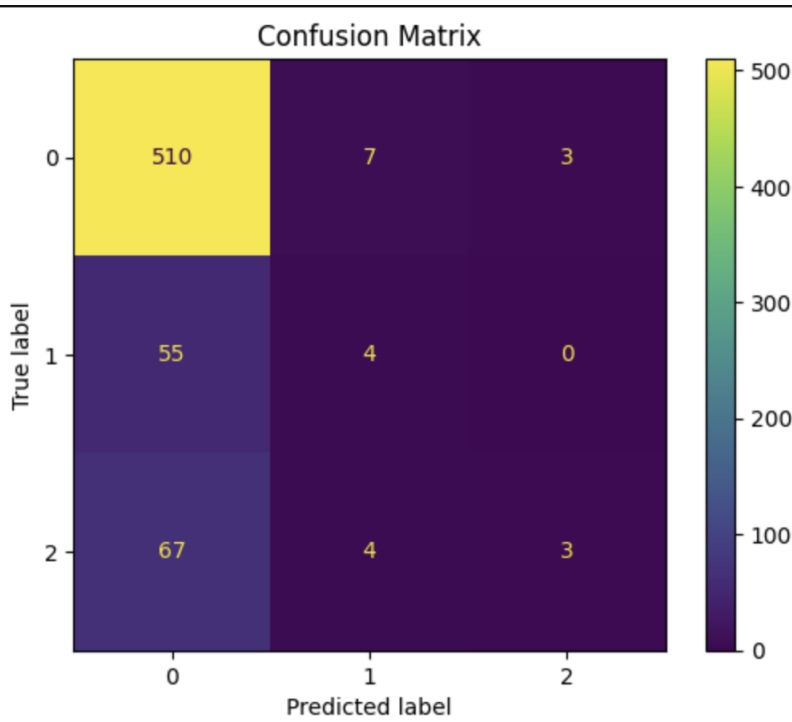


Fig. 2. Confusion matrix of the CNN-LSTM model showing improved classification of sleep onset and wake events.

4.5 Loss and Accuracy Curves

The training and validation curves for both the CNN and CNN-LSTM models, shown in Fig. 3–Fig. 6, provide important insights into their learning behavior and generalization capability.

As observed in Fig. 3, the CNN model exhibits nearly constant training and validation loss across all epochs. The lack of significant decrease in loss indicates that the model fails to effectively learn patterns from the data. Similarly, Fig. 4 shows that both training and validation accuracy remain almost unchanged throughout training. This behavior suggests **underfitting**, where the model is too simple to capture the underlying structure of the time-series data. Although there exists a gap between training and validation performance, the absence of improvement in both metrics confirms that the model is not learning effectively.

In contrast, the CNN-LSTM model demonstrates improved learning dynamics. As shown in Fig. 5, the training loss decreases steadily over epochs, while the validation loss initially increases slightly but subsequently decreases, indicating stabilization and better generalization. This behavior reflects the model's ability to learn meaningful temporal patterns.

Furthermore, Fig. 6 shows that the CNN-LSTM model achieves a gradual increase in training accuracy, while validation accuracy remains relatively stable with minor

fluctuations. The smaller and more stable gap between training and validation curves indicates improved generalization compared to the CNN model.

Overall, the comparison highlights that the CNN model suffers from underfitting due to its inability to capture temporal dependencies, whereas the CNN-LSTM model benefits from sequential learning, resulting in better convergence, improved generalization, and more reliable performance in sleep state detection.

- CNN shows signs of underfitting (high training accuracy, lower validation performance)

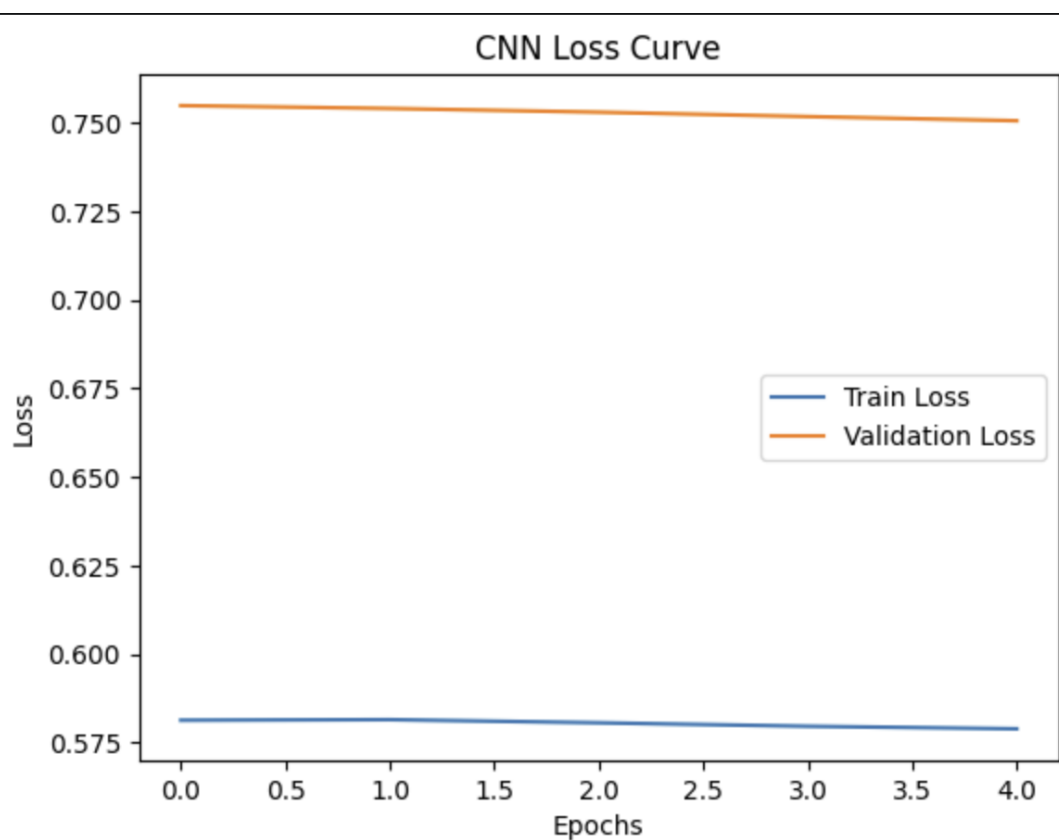


Fig. 3. Training and validation loss curves for the CNN model indicating overfitting behavior.

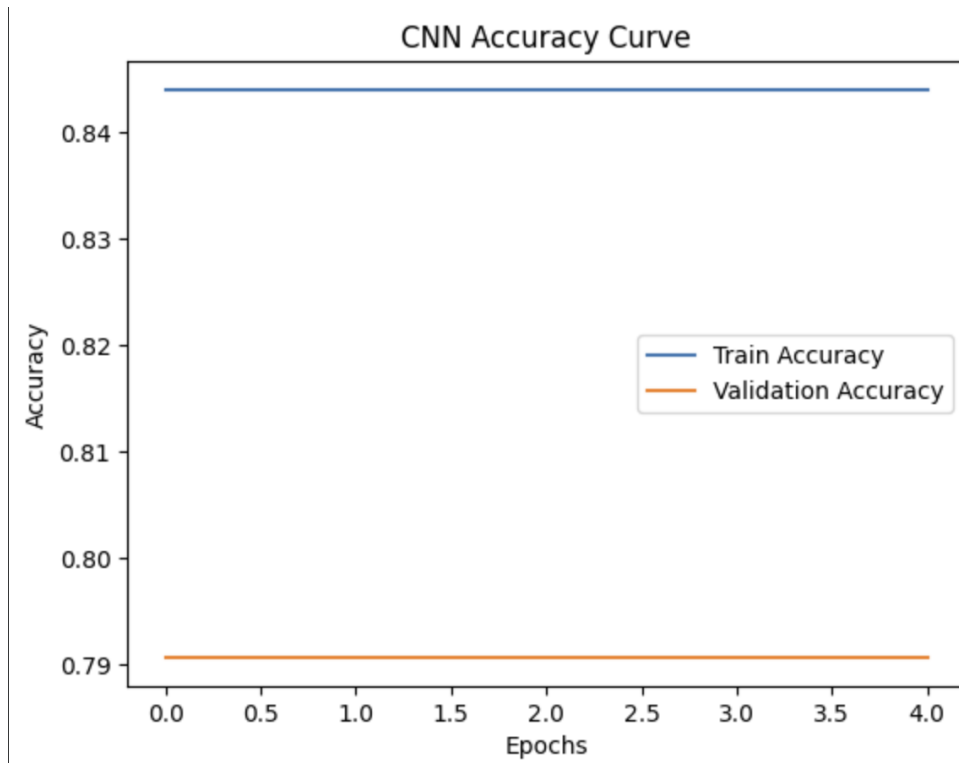


Fig. 4. Training and validation accuracy curves for the CNN model.

- CNN + LSTM demonstrates better generalization

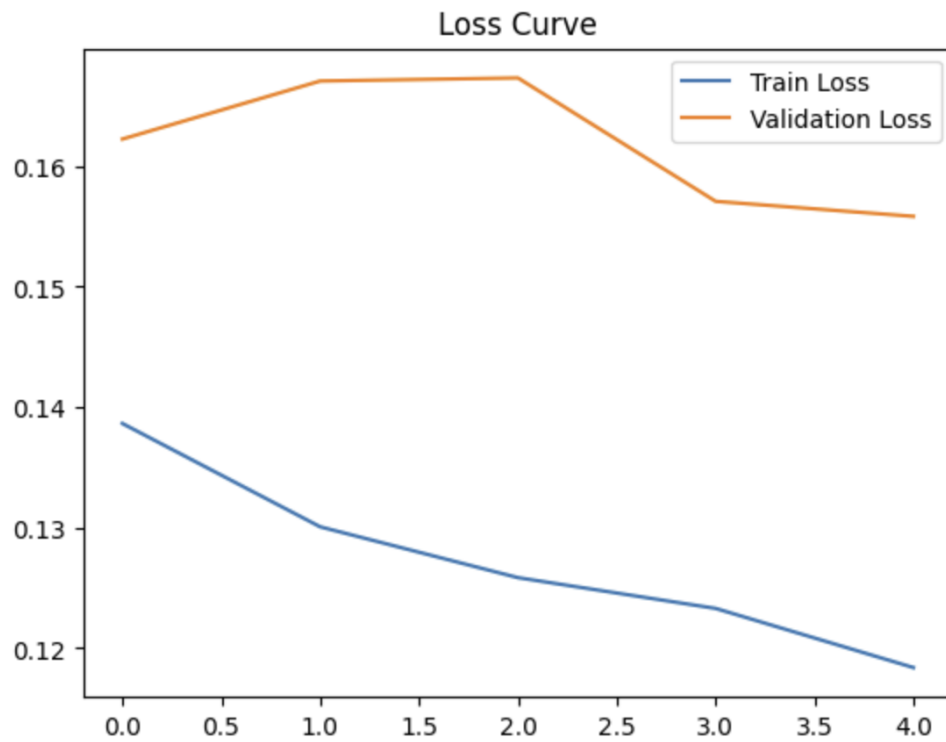


Fig. 5. Training and validation loss curves for the CNN-LSTM model demonstrating improved generalization.

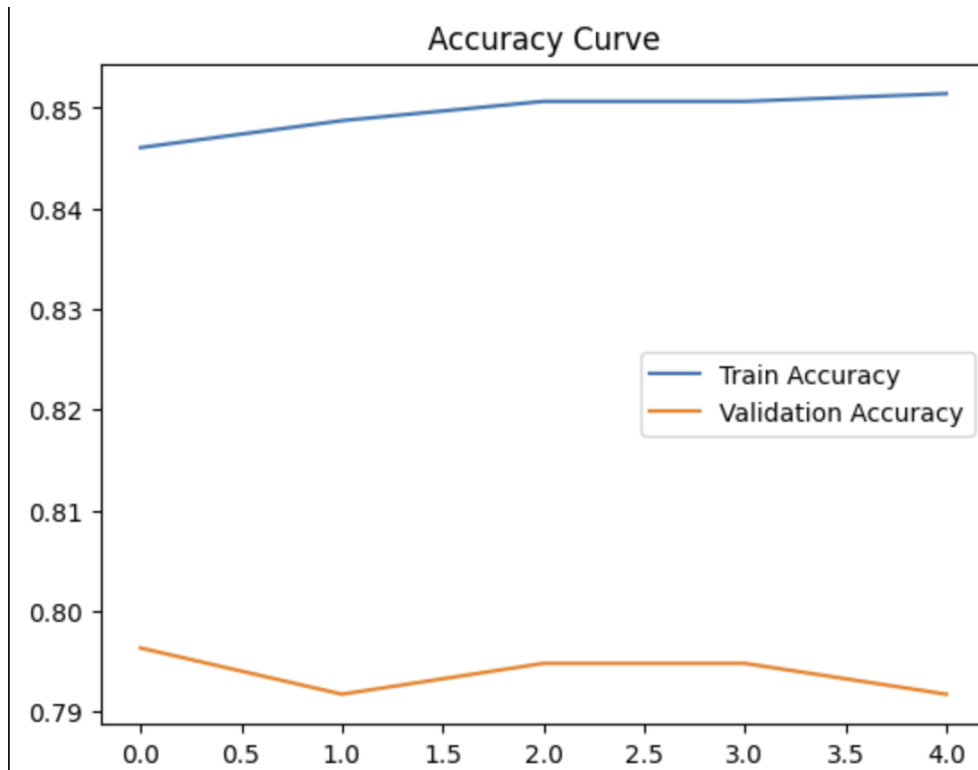


Fig. 6. Training and validation accuracy curves for the CNN-LSTM model.

4.6 Key Observations

- Class imbalance significantly affects model performance
- CNN alone is insufficient for temporal modeling
- Hybrid CNN-LSTM improves event detection
- Post-processing enhances prediction reliability

5. CONCLUSION

This project successfully developed a deep learning-based framework for sleep state detection using accelerometer data. Two models were implemented and evaluated: a baseline CNN model and a hybrid CNN-LSTM model.

The results indicate that while CNN is effective for feature extraction, it lacks the ability to capture temporal dependencies in sequential data. The CNN-LSTM model overcomes this limitation by integrating temporal learning, resulting in improved performance, particularly in detecting sleep onset and wake events.

Despite the improvements, challenges such as class imbalance and overfitting persist. Future work can focus on incorporating advanced architectures such as attention mechanisms, transformer-based models, and improved data balancing techniques.

Overall, this project demonstrates the potential of deep learning for scalable, non-invasive, and real-time sleep monitoring applications.

6. REFERENCES

1. S. Hochreiter and J. Schmidhuber, "Long Short-Term Memory," *Neural Computation*, 1997.
2. Y. LeCun, Y. Bengio, and G. Hinton, "Deep Learning," *Nature*, 2015.
3. A. Krizhevsky et al., "ImageNet Classification with Deep CNNs," 2012.
4. T.-Y. Lin et al., "Focal Loss for Dense Object Detection," *ICCV*, 2017.
5. V. T. van Hees et al., "Estimating sleep parameters using accelerometry," 2015.
6. O. Faust et al., "Deep learning for healthcare applications," 2018.
7. F. Ordóñez and D. Roggen, "Deep CNN for activity recognition," 2016.